ROUTE: Responsive Optimization for Universal Transit Environments

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I. Personal Information Section

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II. Project Proposal Section

Project Title: ROUTE: Responsive Optimization for Universal Transit Environments

Main Area of Interest (choose only the best that applies, put a x mark beside the interest) **:**

_____ Food Safety
_____ Security
__x__ Traffic / Road Congestion
_____ Health
_____ Education
_____ Disaster Mitigation
_____ Entrepreneurship
_____ Agriculture
_____ Environment

Sub-Area(s) of Interest (choose, at most, two (2) interests that sub-categorizes the proposal, put a x mark beside the interest) **:**

_____ Food Safety
_____ Security
_____ Traffic / Road Congestion
_____ Health
_____ Education
_____ Disaster Mitigation
__x__ Entrepreneurship
_____ Agriculture
__x__ Environment

Project Collaborator/s:

1. **Local Government Units (LGUs) and Municipal Franchising Offices:** They manage local transportation concerns and can provide information about TODA terminals, municipal permits, local routes, and fare ordinances.
2. **Department of Tourism (DOT)**: It promotes tourism in the Philippines and provides information about tourist destinations, cultural landmarks, and tourism-related transportation.
3. **Land Transportation Franchising and Regulatory Board (LTFRB):** It regulates public utility vehicles and provides official fare matrices, fare increases, franchises, and transportation regulations.

4. **OpenStreetMap (OSM) Philippines:** It is an open-source mapping platform that provides information about roads, buildings, landmarks, places, and other geographic features.
5. **Transport Cooperatives and TODA Associations:** They represent public transportation operators and can provide information about local routes, vehicle operations, terminal locations, and transportation practices.
6. **Sakay.ph:** It is a Philippine transportation platform that provides commute routes, transportation options, estimated travel times, and directions.
7. **MobilityData (GTFS-Flex Maintainers):** MobilityData supports GTFS and GTFS-Flex standards, which allow transportation information to include flexible routes, on-demand services, and stops without fixed locations.
8. **Transitland (Interline Technologies)**: Transitland is an open transportation data platform that collects and organizes public transit information such as routes, stops, schedules, and transit networks from different sources.
9. **OpenTripPlanner (OTP) Community**: OpenTripPlanner is an open-source journey-planning system that calculates multimodal trips involving walking, buses, trains, and other forms of transportation.
10. **Mapbox:** Mapbox provides digital mapping and navigation services that include interactive maps, street names, turn-by-turn directions, distances, and estimated travel times.

Expected Beneficiaries and Potential Impact of the Proposed Project:

1. **Commuters:** Obtains landmark-guided routes, exact distance in kilometers, and predictable fare bounds. This can help them plan their trips better, avoid getting lost, and know the expected cost before traveling.
2. **Foreign Tourists and Non-Local Travelers:** Protection against *"tourist inflation"* through real-time dual-currency conversion and predicted fare ranges. This can help them understand local transportation costs and avoid confusion when paying in an unfamiliar currency.
3. **Statutory Discount Groups (Students, Seniors, PWDs):** Automatic calculation of 20% legal fare discounts across all transit legs. This allows eligible passengers to easily know the discounted amount they should pay during their journey.
4. **First-Time Visitors in Complex Mega-City Transit (e.g., Tokyo Metro):** Guided through multi-level underground stations, platforms, and exit gates. Clear step-by-step instructions can help them understand transfers and reach the correct platform or exit.
5. **Public Transport Drivers and Operators:** Reduce price arguments and standardize passenger expectations at hail points. Clear fare information can help both passengers and drivers have a better understanding of the expected transportation cost.

Abstract of the Proposal Project:

Public transportation has always been a challenge for commuters and tourists because of informal routes and complicated stations. Some vehicles operate with an unclear schedule, making it difficult for commuters, especially first-timers and tourists, to know the correct fares and roads. This can lead to confusion, causing them to miss their routes, and difficulties when transferring to another vehicle. Furthermore, many transportation routes are not properly illustrated in digital maps, making it harder for passengers to see any pick up routes, terminals, and travel.

To address these problems, this project proposes *Responsive Optimization for Universal Transit Environments* (ROUTE), an AI commuter routing that provides personalized and accessible public guidance. This system will use AI web mining, Natural Language Processing (NLP) and Retrieval-Augmented Generation (RAG), to process travel routes, terminal locations, visual cues, and landmarks based on the available sources. It will incorporate multimodal routes, generating routes like walking and transportations.

ROUTE will have access to a parametric fare calculator that determines the fare based on the distance of the passengers travel as well as statutory passenger discounts. This will also display local fares

equivalent in value to the passenger's currency. Lastly, the visual landmark turn-by-turn GPS will deliver simple instructions using landmarks (e.g stores, waiting areas, footbridges). The system will use transit data to improve route information.

ROUTE is expected to improve navigation, reduce misunderstandings, and make public transportation more accessible and easier for commuters, first-timers, tourists, seniors, and persons with disabilities. This project aims to provide a more understandable and reliable public transit experience.

Additionally, the system includes a Vision-Guided Verification Engine using Optical Character Recognition (OCR) to allow commuters to scan vehicle signboards or landmark signs to confirm their route before boarding.

Problem to be addressed:

Lack of Digital Mapping for Informal Transit: Tricycles, jeepneys, and UV Express vehicles often operate without fixed schedules, mapped routes, or standardized GTFS data, making it difficult for commuters and tourists to locate pickup points and terminals.

Tourist Overcharging and Unclear Fares: Non-local travelers may have difficulty knowing the correct fare because of unfamiliar local fare rules and currencies, which can lead to misunderstandings or unexpected charges.

Complex Station Navigation: First-time commuters may become confused when transferring between different transportation operators, platforms, stations, and exits, especially in large and multi-level transit systems such as the Tokyo Metro.

Proposed Solution to the Problem:

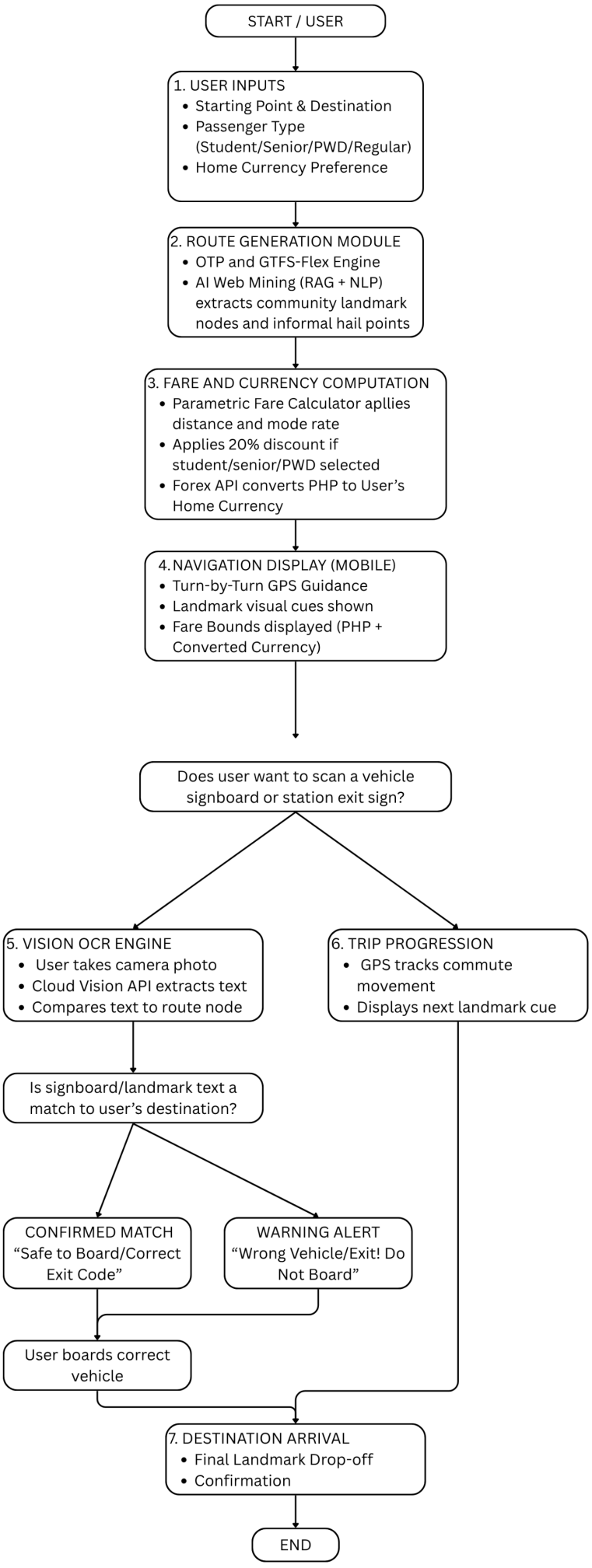
AI Web Mining: Uses Natural Language Processing (NLP) and Retrieval-Augmented Generation (RAG) to extract informal travel directions, terminal locations, pickup points, and visual cues, such as "blue waiting shed" or "behind the market," from public blogs and forums. These details are processed to create spatial landmark nodes for route guidance.

Parametric Fare Calculator: Calculates estimated fare ranges based on travel distance in kilometers or miles, transportation mode (Shared or Special), and applicable legal discount rates.

Dual-Currency Anti-Scam Engine: Automatically converts and displays local fare ranges alongside their equivalent values in the user's home currency, allowing travelers to easily compare and understand transportation costs. For example, a fare of ₱15-₱20 PHP may be displayed as approximately $0.27-$0.36 USD.

Visual Landmark Turn-by-Turn GPS: Provides clear, human-readable navigation instructions using recognizable landmarks and visual cues. For example: *"Walk 150 meters past Mercury Drug, cross the footbridge, and hail a tricycle under the waiting shed"*

Vision-Guided Signboard Verification: Employs Optical Character Recognition (OCR) and Computer Vision to process user-submitted camera photos of vehicle route signboards, instantly verifying if the vehicle passes the user's intended destination before boarding


START / USER
1. USER INPUTS
• Starting Point & Destination
• Passenger Type (Student/Senior/PWD/Regular)
• Home Currency Preference
2. ROUTE GENERATION MODULE
• OTP and GTFS-Flex Engine
• AI Web Mining (RAG + NLP) extracts community landmark nodes and informal hail points
3. FARE AND CURRENCY COMPUTATION
• Parametric Fare Calculator apllies distance and mode rate
• Applies 20% discount if student/senior/PWD selected
• Forex API converts PHP to User's Home Currency
4. NAVIGATION DISPLAY (MOBILE)
• Turn-by-Turn GPS Guidance
• Landmark visual cues shown
• Fare Bounds displayed (PHP + Converted Currency)
Does user want to scan a vehicle signboard or station exit sign?
5. VISION OCR ENGINE
• User takes camera photo
• Cloud Vision API extracts text
• Compares text to route node
6. TRIP PROGRESSION
• GPS tracks commute movement
• Displays next landmark cue
Is signboard/landmark text a match to user's destination?
CONFIRMED MATCH "Safe to Board/Correct Exit Code"
WARNING ALERT "Wrong Vehicle/Exit! Do Not Board"
User boards correct vehicle
7. DESTINATION ARRIVAL
• Final Landmark Drop-off
• Confirmation
END

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Estimated Project Cost, Resource Requirements, and Itemized Budget:

No.	Quantity	Item Description	Unit	Unit Price	Subtotal
1	1	NLP Web Mining API Credits	lot	₱4,500.00	₱4,500.00
2	1	Cloud Vision OCR API Credits	lot	₱2,000.00	₱2,000.00
3	1	Google Maps/Mapbox SDK Credits	lot	₱3,000.00	₱3,000.00
4	1	Live Forex Exchange Rates API Access	lot	₱1,500.00	₱1,500.00
5	6	Firebase Cloud Database and Hosting	mo	₱666.67	₱4,000.00
6	5	Field Mapping and Terminal Transportation	pax	₱700.00	₱3,500.00
7	30	User Testing Mobile Load Incentives	pax	₱100.00	₱3,000.00
				Grand Total	₱21,500.00

Technical Validation and Patentability Check:

Patentability:

No.	Project/Research Title	Problem Statement	Abstract/ Summary of Methodology
1	Landmark-assisted navigation	The need for navigation tools that can selectively utilize database information to provide more precise and relevant guidance based on the user's travel mode and the landmarks of interest.	A computer-implemented method for providing landmark-assisted navigation guidance by selectively utilizing database information includes receiving navigation requests from one or more mobile computing devices, each of the requests being associated with a respective starting point, destination, and travel mode.
2	Landmark navigation	The need for navigation tools that can selectively utilize database information to provide more precise and relevant guidance based on the user's travel mode and the landmarks of interest.	Information of landmark candidates, near a map route, is extracted. A recognizability score for the landmark candidates is assigned based on the extracted information of the landmark candidates.
3	Interactive Landmark-Based Localization	The need for systems that can accurately determine a user's location and orientation using visual landmarks and their	An interactive landmark localization system includes a database of visual landmarks corresponding to

		corresponding geographic information.	prominent physical objects within a geographic area.
4	Multimodal Transit Route Directions	To address the limitation of prior art systems the often recommend routes based on a single mode of transport which may not be the shortest or most cost effective	A method and system for finding multimodal transit route solutions with a computer software program is described that finds the most efficient transit routes based on user preferred modes of transport. The route finding system has four layers that include a presentation layer, an application layer, a component layer and a database layer.
5	Route Search System	Finding the most efficient route between two points, often with the consideration of various factors such as traffic, road conditions, and time constraints. The goal is to provide users with the best possible travel path, whether it be for personal use, business, or transportation.	A route search system that includes a reception unit that receives a search request for a plurality of first routes, that are routes between all the combinations of two of a plurality of points
6	Public and Ordered Transportation Trip Planning	Public and Ordered Transportation Trip Planning involves the need for a system that can efficiently plan and coordinate routes for users traveling with a combination of public transportation and ordered transportation services. This includes the ability to order services such as taxi-cab, ride-share, or micro-mobility options to seamlessly complement fixed-schedule transit.	A method carried out by a computer for coordinating a route comprising a public transportation leg using a vehicle operated by a public transportation service and characterized by an estimated arrival time; and a subsequent ordered transportation leg using a vehicle operated by an ordered transportation service
7	Multi-Modal Fare Calculation , system and apparatus	the lack of a unified, dynamic system capable of calculating cumulative fares across journeys that involve multiple transit modes and multiple mass-transit providers, using disparate fare data sources and multi-segment itineraries.	A computer-implemented method and system are provided for dynamically evaluating fares over multiple itineraries for a journey involving multiple modes of transit and served by one or more mass transit providers.
8	Public Transportation Fare Calculations Based on Geolocation	to automatically determine a passenger's fare in real-time using their actual travel data, while ensuring efficient operation, minimal human intervention, and accurate accounting.	A method for determining transportation fares includes: receiving a plurality of user data points for a user, wherein each user data point corresponds to a geographic location of a user including at least the geographic location and a timestamp
9	Pedestrian Navigation Integrated with Public Transportation	Insufficient Adaptation to Pedestrian Characteristics: Conventional routing systems were typically designed for vehicles, not pedestrians.	A method according to the present disclosure includes receiving information, including a user-desired destination as an end location, from a user-operated computing device and electronically determining a current

			location of a user as a beginning location based on positional information received at the user-operated computing device.
10	Multi-Modal Directions with a Ride Service Segment	A system and method for integrating ride service information directly into a mapping or navigation application, without requiring users to switch to a separate ride service app.	To provide ride services within a mapping application in a client computing device without directing the user to a separate ride service application, the mapping application invokes one or several ride service APIs to access ride service data from various ride service providers.

Related Review of Work:

No.	*Project/Research Title*	*Problem statement*	*Methods or Technology used*	*Strengths*	*Weaknesses*
1	*Learning to Use Transit Services: Adapting to Unfamiliar Transit Travel*	*First-timers face challenges, unfamiliar routes, and psychological anxiety.*	*A qualitative research examining the experience of people who are unfamiliar with commuting*	*Provides insights into the process of habituation for first-timers*	*Does not evaluate the real-time guidance tools.*
2	*Measuring the Impact of Unfamiliar Transit Travel Using a University Access Survey*	*Users who don't know much about public transportation may encounter challenges with navigation during unfamiliar trips.*	*The university used access survey and analysis of first-timers*	*Having navigation, ticketing, and difficulties associated with unfamiliar travel.*	*Doesn't provide a technological solution for problems.*
3	*Is the Use of Informal Public Transport Modes in Developing Countries Habitual? An Empirical Study in Davao City, Philippines*	*Jeepneys and tricycles are commonly used in developing cities, but there is still little research on how passengers choose and get used to these types of transportation.*	*An empirical study of informal transportation in Davao City.*	*Understandin g the informal vehicles such as jeepneys, tricycles, and buses.*	*The study only focuses on Davao City and does not cover changing routes or ways of planning trips.*
4	*SmartTransit PH: An Android App for Public Vehicle Navigation*	*Passengers need information about vehicles, fares, time, distance, and travel*	*Android application for public navigation*	*Gives users useful information like distances, travel time, transfer, and fares.*	*Lacks turn-by-turn navigation for complex, unmapped areas.*
5	*Implementation of Multimodal Public Transportation Route Planner for Metro Manila*	*Metro Manila commuters need a guide when traveling through complicated and hard multimodal transportation networks.*	*OpenTripPlanner and multimodal public transportation route planning.*	*Provide guidance and directions across formal transit networks (e.g trains, bus,*	*Struggles with informal transit networks (e.g tricycles, jeepneys) because of no fixed schedule*

				MRT, LRT)	
6	*A Graph-Based Approach for Efficient, Low-Transfer Routing in Flexible Urban Jeepney Systems*	*The jeepney system operates without a fixed schedule, making it hard for commuters to know when a vehicle will actually arrive.*	*Transfer-Flexible Transit Fare Graph (TFTFGraph), using graph-based routes with distance and fare information.*	*Address the jeepneys transportation fares without relying on the fixed stop.*	*Focuses more on how the routes are calculated than making the routes easy for passengers to understand.*
7	*Personalized Application for Multimodal Route Guidance for Travellers*	*Travelers need updated geographically and personalized public transportation information.*	*Personalized multimodal route-guidance for mobile devices.*	*Gives personalised guidance and considers the multimodal transportation.*	*Mainly focuses on organized transportation systems and does not include informal vehicles (e.g tricycles and jeepneys)*
8	*The Desired Quality of Integrated Multimodal Travel Information in Public Transport*	*Travelers need transportation information that reduces travel time, searching, and effort.*	*Research on the travelers wants and requirements for integrated multimodal travel information.*	*Identifies the important information that reduces any search effort.*	*Does not develop any AI-based system that gives out information.*
9	*En-Route Planning of Multi-Destination Public-Transport Trips Using Smartphones*	*Commuters need real-time and personalised information when planning trips with different destinations.*	*Smartphone system called "Auckland Pioneer", adding real time public information and places.*	*Provides en-route, multi-destination planning and step-by-step guide.*	*Focus more on formal transportations in Western countries and doesn't include informal fares.*
10	*A Multimodal Transport Network Model and Efficient Algorithms for Building Advanced Traveler Information Systems*	*Advanced Traveler Information Systems (ATIS) is made to simplify routes by optimizing speed and distance*	*A multimodal network model considers travel time, transfers, and walking lines.*	*Consider the factors when routed and tested using real transportation networks in Paris.*	*No description and understandable navigation instructions for unfamiliar users.*

My proposal	*ROUTE : Responsive Optimization for Universal Transit Environments*	*Commuters and tourists may struggle with unfamiliar public transportation, informal routes, unclear pick up points, and uncertain fares.*	*AI web mining, NLP, RAG, GIS, multimodal routing, parametric fare calculation, dual-currency conversion, and landmark-based turn-by-turn navigation.*	*Combines route guidance, distance, fare ranges, transportation transfers, pickup locations, landmarks, and personalized instructions in one system.*	*The main research gap is combining these features into one AI-powered guide for people who are unfamiliar with local or international transportation.*

Project Schedule and Timeframe:

	YEAR 2026																
Month	**September**				**October**				**November**				**December**				
Week No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Information Gathering																	
Finalizing Project Title																	
Writing Rationale and Objectives																	
Finalizing Methodology																	
Preparation and Purchasing of Materials																	
System Architecture and Data Schema Design																	
UI/UX Mobile Design																	
API Integration and Database Setup																	
Prototype Building																	
Programming Project Testing and Finalization																	
Data Gathering																	
Data Analysis																	
Writing Final Paper																	